A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 3: Hardware

Subtopic 3.1: Computers and their Components

Concept 3.1.6: Monitoring and Control Systems

Total Questions: 18

Mark Schemes begin on Page 11

Question 1 | Source: 9618_w25_qp_11 (Q9)

9 Describe the differences between a monitoring system and a control system.

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.... [3]

Question 2 | Source: 9618_w25_qp_12 (Q11)

11 An automated system opens doors when a person is detected within 2 metres. The system closes the doors when there is no longer a person within 2 metres.

Identify whether the automated system is an example of a monitoring system or a control system.

Justify your choice.

Monitoring or Control ....

Justification ....

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.... [3]

Question 3 | Source: 9618_s25_qp_11 (Q4.b)

4 A shop installs a new system that allows users to purchase items without going through a manual checkout.

The new system:

- identifies customers when they enter the shop and matches them to their account
- prevents a customer from walking through the automatic barriers if they do not have an account
- automatically detects the items that a customer has taken from a shelf and charges these to the customer's account.

(b) The new system uses sensors to identify the items taken from a shelf.

Identify **one** type of sensor that can be used in this new system.

State how the sensor can be used to identify the items taken from a shelf.

Sensor ....

Use ....

....
[2]

Question 4 | Source: 9618_w24_qp_11 (Q5.a)

5 A security system has both a floodlight (very bright light) and an audio alarm.

The data from multiple sensors is analysed and used to:

- turn on the floodlight
- sound the audio alarm.

Sensors can be used to detect:

- if doors are open
- the external daylight level
- if people are detected within a set distance.

(a) Complete the table to identify the most appropriate type of sensor for each scenario.

Scenario	Type of sensor
A door is open.	
The external daylight level is below a set amount.	
A person is detected within 2 metres.	

[1]

Question 5 | Source: 9618_w24_qp_11 (Q5.c)

5 A security system has both a floodlight (very bright light) and an audio alarm.

The data from multiple sensors is analysed and used to:

- turn on the floodlight
- sound the audio alarm.

Sensors can be used to detect:

- if doors are open
- the external daylight level
- if people are detected within a set distance.

(c) Explain whether the security system is an example of a monitoring system or a control system.

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.... [3]

Question 6 | Source: 9618_w24_qp_12 (Q9.a)

9 A road bridge has a weight limit and a height limit for vehicles. For example, a vehicle must weigh less than 10 000 kg and must have a height of less than 3 m.

The bridge has a warning system. If a vehicle is approaching the bridge and it exceeds one or both limits, a sign displays a warning telling the driver of the vehicle to stop.

(a) The bridge warning system uses sensors to detect if a vehicle exceeds the limits.

Complete the table by identifying **two** different sensors that could be used by the system **and** describe how each sensor is used by the system.

Sensor	Use in bridge warning system
....	....
....	....

[2]

Question 7 | Source: 9618_w24_qp_12 (Q9.b)

9 A road bridge has a weight limit and a height limit for vehicles. For example, a vehicle must weigh less than 10 000 kg and must have a height of less than 3 m.

The bridge has a warning system. If a vehicle is approaching the bridge and it exceeds one or both limits, a sign displays a warning telling the driver of the vehicle to stop.

(b) Explain whether the bridge warning system is an example of a monitoring system or of a control system.

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.... [2]

Question 8 | Source: 9618_w24_qp_13 (Q3)

3 A car has an automatic braking system.

A sensor is used to measure the distance to objects that are in front of the car.

The car automatically brakes if an object is too close to the front of the car or the distance between the car and the object is decreasing too quickly. The closer the object is to the front of the car, the harder the car brakes so that the car slows down more quickly.

Explain the reasons why the automatic braking system of the car is a control system.

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.... [3]

Question 9 | Source: 9618_s24_qp_11 (Q2.b)

2 A video doorbell is attached to the front door of a house. The doorbell uses a motion sensor to detect when a visitor walks in front of the door. When the motion sensor is activated:

- The digital camera in the doorbell starts recording a video.
- A message is transmitted to a smartphone so that the person who lives in the house can watch the video.

The doorbell also has a button that can be pressed. When the button is pressed, a message is transmitted to a smartphone to play the doorbell sound.

The videos are stored on the doorbell's internal secondary storage device and overwritten when the secondary storage device is full.

(b) State whether the video doorbell is a monitoring system or a control system.
Justify your choice.

Monitoring or control system ....

Justification ....

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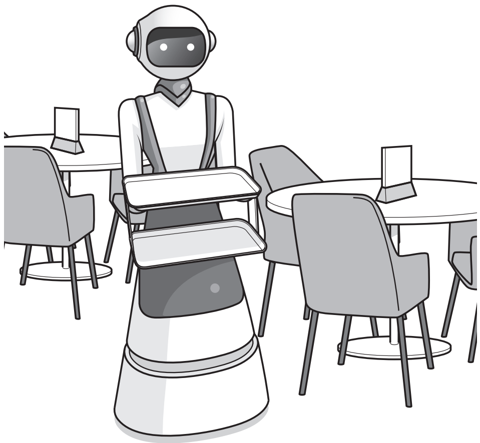
.... [2]

Question 10 | Source: 9618_s24_qp_13 (Q7.a)

7 Robots are used to serve food and drink to customers at a restaurant.



(a) A robot navigates through the restaurant to the table it is serving.

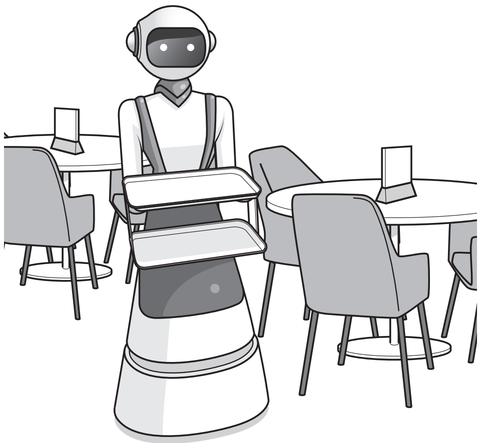
Complete the table by identifying **two** sensors that can be included in the robot **and** the purpose of each sensor in the navigation system.

Sensor	Purpose of sensor in navigation system
....	....
....	....

[2]

Question 11 | Source: 9618_s24_qp_13 (Q7.c)

7 Robots are used to serve food and drink to customers at a restaurant.



(c) The navigation system can be considered an example of a control system.

Describe how feedback is used in a control system.

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.... [2]

Question 12 | Source: 9618_w23_qp_11 (Q7.b)

(b) Describe the purpose of a temperature sensor within the 3D printer.

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.... [2]

Question 13 | Source: 9618_w23_qp_11 (Q9.a)

9 **(a)** Explain the importance of feedback in a control system.

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.... [2]

Question 14 | Source: 9618_w23_qp_12 (Q1.a)

1 A factory makes chocolate bars.

The factory uses a conveyor belt that moves the products from one stage of production to the next stage.

(a) An automated system counts the number of chocolate bars made at the end of production.

The system includes a sensor positioned above the conveyor belt.

Identify **one** appropriate type of sensor that can be used.

.... [1]

Question 15 | Source: 9618_w23_qp_12 (Q1.b)

1 A factory makes chocolate bars.

The factory uses a conveyor belt that moves the products from one stage of production to the next stage.

(b) A second automated system removes chocolate bars with an incorrect weight from the production line.

Describe the role of an **actuator** in this second system.

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.... [2]

Question 16 | Source: 9618_w22_qp_13 (Q10.a)

10 **(a)** Explain the importance of feedback in a control system.

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.... [3]

Question 17 | Source: 9618_w22_qp_13 (Q10.b.i)

(b) **(i)** Identify **one** sensor that could be used in a car alarm system.

Justify your choice.

Sensor ....

Justification ....

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[2]

Question 18 | Source: 9618_s21_qp_11 (Q5.c)

5 Kiara has a washing machine and a refrigerator.

(c) The temperature in her refrigerator must be kept between 4 and 6 degrees Celsius.

The microprocessor in the refrigerator turns on the cooling if the temperature is too high, and turns off the cooling if the temperature is too low.

Explain why the system in the refrigerator is a control and not a monitoring system.

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.... [2]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_11 (Q9)

9	**1 mark** per bullet point, **max 3** marks • Monitoring systems do not take any action whereas control systems act autonomously to change the environment if the values are out of a prescribed range • Control systems use actuators monitoring systems do not have any actuators • Control systems make use of feedback monitoring systems do not make use of feedback • The output from a monitoring system does not affect the subsequent input whereas output from a control system affects the next input.	**3**

Mark Scheme for Question 2 | Source: 9618_w25_ms_12 (Q11)

11	**1 mark** per bullet point, **max 3** marks Identification: Control system (no mark) Justification: • The system uses an actuator to open or close the doors • The output / opening the doors allows people to move, which changes the input to the sensor • The system acts autonomously on the feedback from a sensor	**3**

Mark Scheme for Question 3 | Source: 9618_s25_ms_11 (Q4.b)

4(b)	**1 mark** for sensor, **1 mark** for corresponding use e.g.	**2**

Sensor	Use
Pressure	Detects when the pressure of an item is removed from a shelf / put back on a shelf
Infrared	Detects when the beam is broken for an item removed / added

Mark Scheme for Question 4 | Source: 9618_w24_ms_11 (Q5.a)

5(a)	**1 mark** for all 3 appropriate sensors <table><tr><th>Scenario</th><th>Type of sensor</th></tr><tr><td>A door is open.</td><td>Pressure sensor // Infra-red</td></tr><tr><td>The external daylight-level is below a set amount.</td><td>Light sensor</td></tr><tr><td>A person is detected within 2 metres.</td><td>Infra-red</td></tr></table>	**1**

Mark Scheme for Question 5 | Source: 9618_w24_ms_11 (Q5.c)

5(c)	**1 mark** for each bullet point (**max 3**) for a correct justification No marks for the identification of the system **Monitoring system** • There is no use of feedback // The light and the alarm are just warnings • The output of the floodlight or the audio alarm does not affect the input of data from the sensors • The system does not have any actuators	**3**

Mark Scheme for Question 6 | Source: 9618_w24_ms_12 (Q9.a)

9(a)	**1 mark** for sensor **and** its use in this system • Infra-red sensor • Measure / check the height of the vehicle • Pressure sensor • Measure / check the weight of the vehicle	**2**

Mark Scheme for Question 7 | Source: 9618_w24_ms_12 (Q9.b)

9(b)	**1 mark** for each bullet point (**max 2**) for a correct justification No marks for the identification of the system Monitoring system • Because there is no use of feedback // the warning sign is only an indicator • ... the output of the turning on of the sign does not affect the input of data from the sensors • The system does not have any actuators	**2**

Mark Scheme for Question 8 | Source: 9618_w24_ms_13 (Q3)

3	**1 mark** for each bullet point (**max 3**) • The system uses feedback • The system acts independently • Input data causes the car to brake • … braking decreases the/changes the distance to the object • … this new distance is used to determine whether/how hard the car needs to brake	**3**

Mark Scheme for Question 9 | Source: 9618_s24_ms_11 (Q2.b)

2(b)	No mark for identification of monitoring or control **1 mark** each to **max 2** for justification: Monitoring: • The turning on of the digital camera does not affect the input to the sensor/button • The transmission of the data/video does not affect the input to the sensor/button • The ringing of the doorbell does not affect the input to the button Control: • Video doorbell does not only store the values from the motion sensor • The data is processed, generating a signal to start the digital camera recording • Button pressed/motion detected causes a signal to be sent over a network to the smartphone	**2**

Mark Scheme for Question 10 | Source: 9618_s24_ms_13 (Q7.a)

7(a)	**1 mark** for sensor **and** matching purpose to **max 2**:	**2**

Sensor	**Purpose of sensor in navigation system**
Pressure	To detect if a table or other obstacle has been hit // to detect when food is put on/taken off the tray so it can move on
Infra-red	To detect if there is an obstacle in the way // to indicate that it has reached the desired table
Sound	To detect if someone is speaking so that it can use AI to decipher the speech and whether the robot is required to stop

Mark Scheme for Question 11 | Source: 9618_s24_ms_13 (Q7.c)

7(c)	**1 mark** each to **max 2**: • Feedback ensures that a system operates within set criteria / constraints • ... by enabling system output to affect subsequent system input • ... thus allowing conditions to be **automatically** adjusted	**2**

Mark Scheme for Question 12 | Source: 9618_w23_ms_11 (Q7.b)

7(b)	**1 mark** for each bullet point (**max 2**) • To prevent **overheating** // ensure material is hot enough • ...by identifying the temperature of the object (being printed) • ...by identifying the temperature of the material being used	**2**

Mark Scheme for Question 13 | Source: 9618_w23_ms_11 (Q9.a)

9(a)	**One mark** for each bullet point (**max 2**) • Feedback ensures that a system operates within set criteria / constraints • ...by enabling system output to affect (subsequent) system input • ...thus allowing conditions to be <u>automatically</u> adjusted	**2**

Mark Scheme for Question 14 | Source: 9618_w23_ms_12 (Q1.a)

1(a)	**1 mark** for: • infra-red / proximity (sensor)	**1**

Mark Scheme for Question 15 | Source: 9618_w23_ms_12 (Q1.b)

1(b)	**1 mark** for each bullet point (**max 2**) • Actuator generates a signal / causes an action / converts electrical energy into a mechanical force • ... to push an arm // to open a trap door // to pick up the chocolate bar with the incorrect weight	**2**

Mark Scheme for Question 16 | Source: 9618_w22_ms_13 (Q10.a)

10(a)	**1 mark** for each bullet point: • to ensure the system operates with the given criteria • ... by enabling system output to affect subsequent system input • ... thus allowing conditions to be <u>automatically</u> adjusted	**3**

Mark Scheme for Question 17 | Source: 9618_w22_ms_13 (Q10.b.i)

10(b)(i)	**1 mark** for identification of a suitable sensor **1 mark** for corresponding justification Example: • sound sensor • if a sound occurs inside the car the alarm is activated • infra-red sensor • senses the heat of person in the car / infra-red beams are broken • pressure sensor • an intruder sits in the driver's seat	**2**

Mark Scheme for Question 18 | Source: 9618_s21_ms_11 (Q5.c)

5(c)	**1 mark** per bullet point • The system uses feedback • The system causes the temperature to change // produces an action	**2**